by Kunal Sir

What is Type Casting in Java?

- Type Casting is the process of converting the value of a single data type (such as integer [int], float or double) into another data type.

- This conversion is done either automatically or manually.

- The compiler performs the automatic conversion, and a programmer does the manual conversion.

Syntax of Type Casting:

```
<datatype> variablename = (<datatype>) value;
```

Types of Type Casting in Java:

There are two type of type casting in java which is given below-

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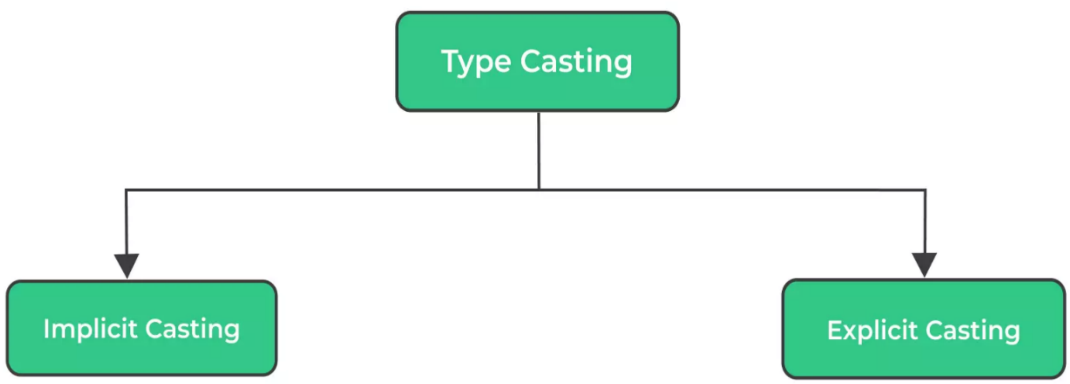
- **Implicit / Widening / Automatic Type Casting**
- **Explicit / Narrowing / Manual Type Casting**

Java Type Casting



Types of Java Casting Or Conversion

1. Implicit Type Casting:

- Implicit type casting is the process of converting a lower data type to a higher data type.
- It is also referred to as Widening conversion or casting

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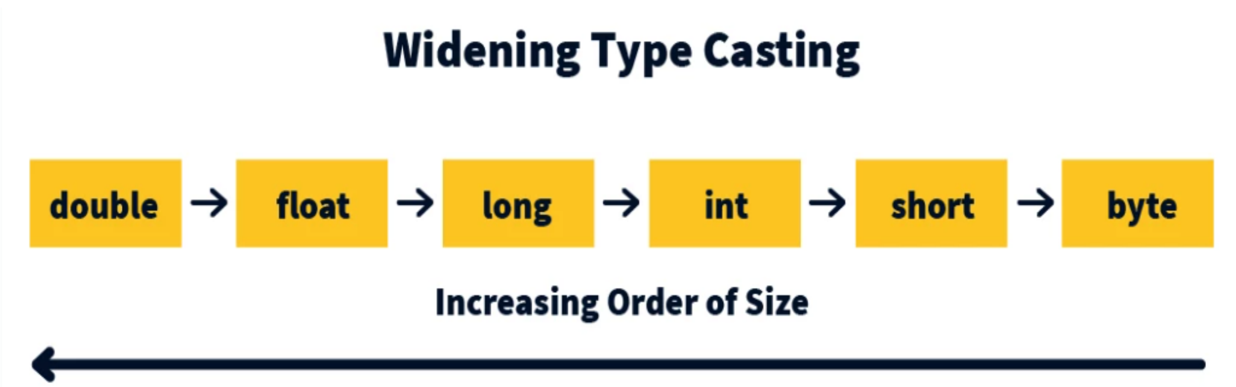
down. This process is performed automatically and is safe, as there is no risk of the data loss.



2. Explicit Type Casting:

- Explicit type casting is the process of reducing a larger data type to a smaller one.
- Other names for this are casting up or narrowing type casting in java.
- It does not happen on its own.
- If we do not do this explicitly, we will get a compile time error.
- The explicit type casting is unsafe because data loss might occur due to the smaller range of allowed values of the lower data type.

Importance the Type Casting in Java:

Type conversion and casting in java bear huge importance. Below is the most significant reason to use type casting or type conversion -

- **Preventing Data Loss: -** When we change data from one type to another, we need to ensure that we do not lose any valuable information. Type casting helps us convert data safely while keeping its integrity intact.

- **Making Operations Work: -** Imagine trying to mix a glass of water with a cup of sugar. It just would not blend! Similarly, some operations can not work with different data types. Types casting makes these operations possible by making data types compatible.

- **Efficient Memory Use: -** Type casting and type conversion in java help manage memory better. Sometimes, a larger data type might be used for a smaller value. This can waste memory. Type casting lets us use the right size for the right data.

- **Handling User Input: -** When we get input from users, it is usually in text form. But we need numbers for calculations. Type casting helps convert this text into numbers for proper processing.

- **Better Programming Control: -** Java is a strongly-typed language, which means it is strict about data types. Types casting and type conversion in java allow programmers to take control and ensure the right data type is used at the right time.

Difference between Implicit & Explicit Type Casting in Java: -

There are following difference between implicit & explicit type casting which are given below-

Implicit Type Cast	Explicit Type Cast
Implicit type casting done automatically by jvm	Explicit type casting done manually by programmer
It is also known as widening type casting	It is also known as narrowing type casting
Implicit type casting does not require any special syntax	Explicit type casting requires the syntax for casting
In implicit type cast there is no loss of data during conversion	In explicit type cast data loss may or may not be takes place